LinStabPy: compressible stability and resolvent analysis

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Abstract

We present LinStabPy, an open Python framework for compressible linear stability and resolvent analysis built around a shared discrete operator for supplied base or mean flows. The discretization employs summation-by-parts finite differences on curvilinear grids; the resulting sparse systems are solved through PETSc/SLEPc. A software layer reuses the same assembled operator across temporal stability, spatial stability, and resolvent workflows. Independent operator checks, fidelity assessment against the reference MATLAB implementation, and validation in an oblique shock-wave/boundary-layer-interaction setting are collected in the Supplementary Material. The implementation is released as the Python package `linstabpy`.

Keywords: compressible stability; resolvent analysis; linearized Navier–Stokes; sparse solvers; PETSc; SLEPc

1 Introduction

Linear stability and resolvent analyses remain standard tools for compressible-flow problems once a reference state is already available from simulation, experiment, or a separate base-flow solver [1–4]. In this setting the central object is the discrete linearized operator, which—rather than the base-flow generator itself—supports modal, nonmodal, and frequency-domain analyses. The practical bottleneck is therefore twofold. One needs a trustworthy compressible discretization, and one needs a solver stack that remains usable when the resulting sparse systems become large.

LinStabPy builds on two public repositories: the `LinStab2D` repository [5, 6] and the `resolvent4py` repository [7–9]. The former provides the compressible formulation on structured and curvilinear grids, including summation-by-parts finite differences, boundary-condition recipes, and resolvent analysis. The latter introduces reusable operator abstractions for large-scale linear analyses. LinStabPy combines these layers: a compressible discrete operator for supplied flows, an analysis interface written in terms of that operator, and interchangeable PETSc/SLEPc solver backends for direct and iterative linear-algebra workflows.

The surrounding software ecosystem is expanding. Packages such as `BROADCAST` [10], `dNami` [11], and `LightKrylov` [12] reflect growing open-source activity around operator-based workflows and Krylov-based numerics; see Frantz et al. [13] for a review of Krylov methods in fluid-dynamics applications.

LinStabPy addresses a narrower workflow than these broader packages. Many users can obtain a nonlinear base or mean flow but lack a ready-to-use compressible linear operator with the needed thermodynamic, curvilinear, and boundary-condition support. LinStabPy fills that gap: the base or mean flow is obtained from the user's own CFD or data-generation environment, and the package discretizes the corresponding linearized operator for prototyping and production-scale stability/resolvent calculations.

The main contribution is the framework itself: rather than reporting new flow-physics findings, we provide a computational tool that turns supplied compressible reference states into a discrete operator reusable across temporal stability, spatial stability, and resolvent analyses. The detailed formulation, verification protocol, and benchmark results are collected in the Supplementary Material.

2 Framework

The framework takes as input a compressible base or mean flow together with the mesh, finite-difference order, boundary-condition recipe, and filter choices. Together, these define a fully discrete linearized operator. Temporal stability, spatial stability, and resolvent analysis then act on that shared operator or on derived shifted/block variants, while the sparse linear algebra is handled through PETSc/SLEPc [14, 15] with SLEPc

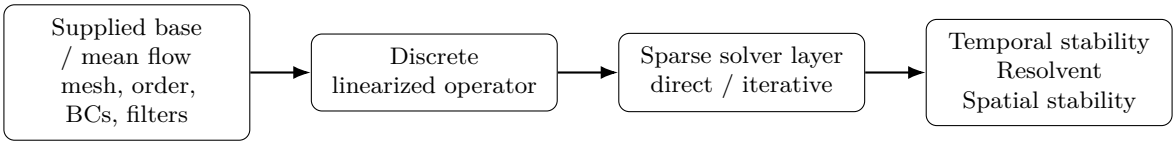


Figure 1: LinStabPy framework overview. User-supplied inputs define a discrete linearized operator, which feeds stability and resolvent workflows through a direct or iterative sparse solver layer.

Krylov–Schur plus shift-invert and MUMPS direct factorization [16] for eigenvalue problems, and either direct or iterative Krylov/preconditioning routes [17, 18] for resolvent analyses.

Because the operator is assembled explicitly as a sparse matrix, the current eigenvalue workflow uses SLEPc Krylov–Schur as the outer eigensolver together with a shift-invert transformation, while each shifted linear system is solved directly by sparse LU factorization with MUMPS. This is distinct from the time-stepper approach [19, 20], in which the linearized operator is never formed and Krylov methods serve as the *outer* eigensolver acting on propagator snapshots; see Frantz et al. [13] for a review. The two approaches use "Krylov methods" in different roles (outer iteration on the propagator versus inner iteration on a shifted system), and the distinction matters for solver selection: in the current matrix-forming eigenvalue workflow the outer method is Krylov–Schur, but the inner shifted solves are direct rather than iterative, whereas the time-stepper outer loop avoids inversion altogether. The Supplementary Material (Section S6) discusses this point in detail.

The framework separates physical assembly from operator analysis: changing the finite-difference order, coordinate mapping, or boundary recipe affects only the operator-construction step, not the downstream analysis logic. This design builds on the compressible formulation of Martini and Schmidt [6] while providing a reusable software layer for current and future operator-based methods, including the harmonic-resolvent extensions discussed in Padovan et al. [21], Padovan and Rowley [22], Farghadan et al. [23].

3 Verification and scope

The Supplementary Material presents verification evidence in three parts: two independent checks of the assembled discrete operator, fidelity assessment through the five shipped Martini examples [6], and validation in the oblique SWBLI setting of Robinet [24] and Song and Hao [25]. It also contains the compressible linearized Navier–Stokes derivation, curvilinear and SBP discretization details, solver configuration, the discrete Chu energy norm, and the harmonic-resolvent extension.

4 Code availability

LinStabPy is available at `https://github.com/ricardofrantz/linstabpy` as the Python package `linstabpy`; release DOI and snapshot metadata will be added in the final submission package.

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